RACE 2 ZERO: OUTBOARD MOTORS IN THE GALAPAGOS ISLANDS

Stuff

Race to Zero

Green Microgrid for Marina Charging Stations on Galapagos Islands

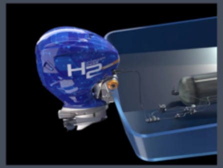
PURDUE UNIVERSITY | Vertically Integrated Projects

Executive Summary

Drawing inspiration from the SUMMIT 2023 in the Galapagos, this project aims to decarbonize the Galapagos Islands with renewable energy-powered microgrids providing infrastructure for hydrogen and electric-powered outboard motors.

The project has four main goals:
- Implement one or both of green outboard motors, along with an associated green grid.
- Create a positive environmental impact on the local marine life.
- Increase tourist and customer satisfaction with the implementation of less noisy and smelly engines.
- Become the main self-sustaining engine type used by the island's boating companies.

Yamaha H2 Outboard Motors



Mercury Electric Outboard Motors

Green Microgrid

With alternatives to petrol, the proper infrastructure has to be put in place. The company Kaizen provides a system that provides the capability to recharge electric motors and refuel hydrogen combustion engines with its built in electrolyzer.

- Generates hydrogen using methanol reforming
  - Stores energy for later conversion to electricity or direct hydrogen provision
  - AC38 MWh per day per unit
- No capital cost - provided under "as service" contract

Kaizen Green Energy:
Propoposed Microgrid Solution

Offboard Engines

Yamaha H2 Outboard Motors

Environmental Sustainability
- Zero emissions
- Water is the only waste product

Enhanced Tourism Experience:
- Offers greater horsepower with ratings of 425 and 450

Economic feasibility:
- More expensive to both produce than electric power and gasoline
- More expensive to the consumer when refueling
- Saves money in long-term maintenance as opposed to gasoline counterpart

Mercury Electric Outboard Motors

Environmental Sustainability
- Zero emissions

Enhanced Tourism Experience
- Quieter and smoother rides for tourists
- Attracts environmentally conscious tourists

Expanded Access to Protected Areas
- Allows access to previously inaccessible areas
- Promotes responsible tourism practices

Cost Savings and Economic Feasibility
- Lower operational costs and reduced maintenance expenses

SCAN ME

Business/Policy model

- Seek government support to accelerate adoption
  - Products tie well into Galapagos Islands' energy transition initiatives
- Use "cash for clunkers" trade in model to rapidly phase out existing gasoline engines
  - Will likely require government subsidies
- Provide supporting infrastructure (microgrids) at marinas to support energy transition

Benefits VS gasoline:
- Reduces pollution/risk of oil spills
- Lower cost – cheaper than imported gasoline
- On-demand fuel generation
- Increased self sufficiency

Policy recommendations:
- Accelerate research into alternative fuels
- Discourage the use of fossil fuel based electricity (e.g. taxes, quotas)
- Subsidize expansion of renewable energies
- Provide incentives/benefits to firms that use alternative fuel marine engines

Business Plan

Business Description:

- Introducing electric motors and hydrogen combustion engines to the Galapagos Islands
- Targeting sustainability and eco-friendliness in the Galapagos Islands
- Offering a cleaner, more efficient, and quieter alternative to traditional engines

Market Analysis:

- Galapagos tourism revenue: $200 million
- Current reliance on gasoline-powered boats

Product Strategy:

- Two product lines: hydrogen combustion outboards and inboard fuel cell boats.
- Advantages: Noise reduction, fume elimination, efficiency improvement, smoother ride.
- Simplified fuel supply chain, eco-friendly initiatives.

Operating Plan:

- Partnerships with Yamaha and Mercury for supplies
- Collaboration with local governments for trade-in programs and environmental organizations

Tech Assessment

- Benefits to consumers
  - Increased tourism for more pleasant boat rides
- Lowers long-term business costs
  - Variable costs with hydrogen and electric motors are lower than gas-powered engines
  - Electric motors have extremely low recharging costs
- A government supported and subsidized programs
- Companies with large market shares are making the same switch to zero-emission motors
  - The high demand of products from companies like Rivian and Tesla shows the consumers' willingness to switch to electric or hydrogen-powered motors, making the plan economically feasible
  - Reputable companies like Yamaha have similar plans to switch to an outboard motor with no carbon emissions

Website

ARTICLES AND SOURCES

Mercury Avator™ Electric Outboard Motors | Mercury Marine

Electric Outboard Motors - Avator electric outboards are as kind to the environment as they are simple to use and fun to drive. Clean, quiet power that moves you.

yamahaoutboards.com/newsroom/company-news/yamaha-develops-hydrogen-fuel-system-with-roush-and-regulator-marine-hydrogen-outboard-unveiled-at

Yamaha's world-first hydrogen outboard unveiled on prototype boat

Yamaha previously announced it would show its prototype hydrogen-combustion outboard motor at the 2024 Miami International Boat Show, and it's gone the extra mile in bringing a prototype boat and fuel delivery system to go along with it. Yamaha calls its new hydrogen V8 outboard a world first for...

Outboard Engine Comparison Matrix | BoatTEST

KAIZEN Clean Energy

OUTERENVY OUTBOARD MOTOR COVERS

between 2 stroke and 4 stroke outboard motors is beginning to close. That being said, there are still differences between the two. It

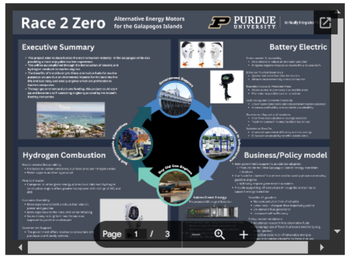
RACE TO ZERO

WELCOME

We are a team of undergraduate students from Purdue's Engineering College and Business College working on research into emerging hydrogen technologies. This spring 2024 semester we focused on researching how to decarbonize the Galapagos Islands.

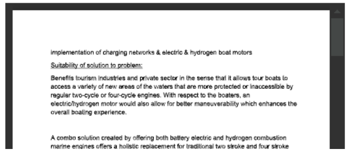
WHAT WE DID THIS SEMESTER

Final Poster Presentation Options



Poster Presentation Draft

Technical Assessment Draft



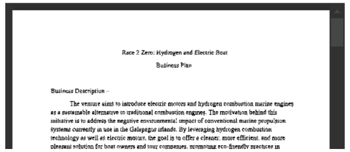
implementation of charging networks & electric & hydrogen boat motors

Suitability of solution to problem:

Benefits tourism industries and private sector in the sense that it allows tour boats to access a variety of new areas of the waters that are more protected or inaccessible by regular two-cycle or four-cycle engines. With respect to the boaters, an electric/hydrogen motor would also allow for better maneuverability which enhances the overall boating experience.

A combo solution created by offering both battery electric and hydrogen combustion marine engines offers a holistic replacement for traditional two stroke and four stroke

Business Plan Draft



Race 2 Zero: Hydrogen and Electric Boat

Business Plan

Business Description –

The venture aims to introduce electric motors and hydrogen combustion marine engines as a sustainable alternative to traditional combustion engines. The motivation behind this initiative is to address the negative environmental impact of conventional marine propulsion systems currently in use in the Galapagos islands. By leveraging hydrogen combustion technology as well as electric motors, the goal is to offer a cleaner, more efficient, and more pleasant solution for boat owners and tour companies, promoting eco-friendly practices in